

22/09

Now : Want to Print Like this

```
for i in range(1, 6, 1):
```

```
    for j in range(5, i, -1):
```

```
        Print(" ", end = " ")
```

```
    for k in range(1, i+1, 1):
```

```
        Print("* ", end = " ")
```

Print("\n")

here we
used two
Nested
for loop

```

- - - - *
- - - *  *
- - *  *  *
- *  *  *  *
*  *  *  *  *
```

```

for i in range(1, 6, 1):
    # this loop calculating the space
    for j in range(5, i, -1):
        Print(" ", end=" ")
    # this loop was printing the stars
    for k in range(1, i+1, 1):
        Print("*", end=" ")

```

o/p:- Print()

```

      *
     **
    ***
   ****
  *****

```

Now i want to Print like this

```

      *
     **
    ***
   ****
  *****
  *****
  ****
   ***
    **
     *

```

up to this program

Next this program
but we create
both one

logic

i = 1, 2, 3, 4, 5

j = 1

j = (5, 1, -1) =

5, 4, 3, 2 → 4 spaces Print

k (1, 2) → k = 1

Print * → 1 star

i = 2

(5, 2, -1) → j = 5, 4, 3

3 spaces

k (1, 3) = 1, 2 → Print **

2 stars

i = 3

(5, 3, -1) → * no iteration

0 spaces

k (1, 4) k = 1, 2, 3, 4, 5

Print *****

5 stars

```

num = int(input("enter the row number: "))

```

row = 1

```

while (row <= num):

```

```

    col = 1

```

```

    while (col <= row):

```

```

        Print("*", end=" ")

```

} Print \n

col = col + 1

Print ()

row = row + 1

row = 1

while (row <= 5):

col = 4

while (col >= row):

Print ("*", end = " ")

col = col - 1

Print ()

row = row + 1

Print Δ/
Part

logic: first Part

Outer loop (row) → controls how many rows to Print. Runs from 1 to num.

Inner loop (col) → Prints star (*) in each row. Number of stars = value of row

Second Part

Outer loop (row) → controls rows. ~~Runs~~ Runs from 1 to 5

Inner loop (col) starts from 4 and goes downwards to row

• So first row prints 4 stars, second prints 3, ...
untill last row prints 0.

Step by Step

row = 1 → col = 4, 3, 2, 1 → 4 stars

row = 2 → col = 4, 3, 2 → 3 stars

row = 3 → col = 4, 3 → 2 stars

row = 4 → col = 4 → 1 star

row = 5 → inner loop skipped → 0 stars

```

for i in range(1, 8, 1):
    for j in range(1, i+1, 1):
        Print("*", end=" ")
    Print()

```

```

for x in range(1, 5, 1):
    for y in range(5, x, -1):
        Print("*", end=" ")
    Print()

```

O/P

```

*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

Now I want to Print like this

Spaces b/w

```

* - - - - *
* * - - - - * *
* * * - - - * * *
* * * * - - * * * *
* * * * * * * *
* * * * - - * * * *
* * * * - - * * *
* * - - - - * *
* - - - - - *

```

```

*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

```

for i in range(1, 6, 1):
    Print("* " * i, end=" ")
    Print(" " * (2 * (5 - i)), end=" ")
    Print("* " * i, end=" ")
    Print()

```

```

for j in range(4, 0, -1):
    Print("* " * j, end=" ")
    Print(" " * (2 * (5 - j)), end=" ")
    Print("* " * j, end=" ")
    Print()

```

logic

The pattern has a top half (first for i in range $(1, 5)$) and a bottom half (second for j in range $(4, 0, -1)$). Each line consists of three parts printed consecutively

- 1) Left block of stars: "*" * count
- 2) Middle block of spaces: " " * $(2 * (5 - \text{count}))$
- 3) Right block of stars: same as the left block
- The middle block width is $2 * (5 - 1)$ so the total width stays symmetric
- The second loop uses range $(4, 0, -1)$ (4 down to 1) so the center row ($i = 5$) is not repeated.

Print func details

- Print ("*" * i , end=" ") Print left stars without newline
- Print (" " * $(2 * (5 - i))$, end=" ") Print middle spaces without newline.
- Print ("*" * i , end=" ") prints right stars and a single trailing space (that's why you see a trailing space before the newline)
- Print("\n") Prints a newline.

iteration by iterat (top half)

$i = 1$

left stars: $1 \rightarrow *$

middle spaces: $2 * (5 - 1) = 8 \rightarrow " "$

right stars: $1 \rightarrow *$

line: * *

$i = 2$

LS: $2 \rightarrow **$

MS: $2 * (5 - 2) = 6 \rightarrow " "$

RS: $2 \rightarrow **$

line: ** **

⋮

i = 5

LS : 5 + *****

MS : 2 * (5-5) = 0 + "

RS : 5 + *****

line : *****

Iterat by Iterat (bottom half)

Cj goes 4 + 1

• j = 4 → *****

• j = 3 → *****

• j = 2 → *****

• j = 1 → *****

mirror
image of top half,
excluding the central
i = 5 line

why 2 * (5-i) ?

Using 2 * (5-i) makes the middle block shrink by 2 spaces
each time i increases by 1. — not keeps the left and right
same - blocks moving together symmetrically until they
meet at i = 5

Instead of Writing 3 Print Statement Now I
Can Write in a Single Print Statement.

for i in range(1, 5):

Print (" " * i + " " * (2 * (5-i)) + " " * i)

for j in range(4, 0, -1):

Print (" " * j + " " * (2 * (5-j)) + " " * j)
